ic_autolab round 19 (generic track): report

Protocol: PROTOCOL.md, frozen 2026-11-04T06:48:30Z at git `ff7401cbf7c5fe5659e1b6262a7128965a33ba33`; route `local`. Counted GAE only (deterministic, host independent); no wall-time claim; development loop, no validation-gate row. Loop home: `../`; ledger: `../ledger.json`.

Claim record: `PCC-20261104-997808` (`python3 tools/pcc/pcc.py standing PCC-20261104-997808`). Until a signed replay and recompute exist, this round's outcome is an unverified assertion [pcc:unverified].

Outcome

k	class	sel	sel / inc at w = 0 [95%]	w = 0.1	w = 1	band	S sel (w = 0 / 0.1 / 1)	S inc	S rho-neg	distance sel / rho
32	**improvement**	wsum-b20-n3	0.350 [0.329, 0.373]	0.360 [0.337, 0.386]	0.416 [0.377, 0.458]	sel < inc	2.937 / 3.213 / 5.697	8.384 / 8.917 / 13.710	0.939 / 1.032 / 1.862	3.1x / 3.1x / 3.1x

Classes follow the protocol's decision rule. Distance to rho is the distance remaining, not a gate. The reasons per size:

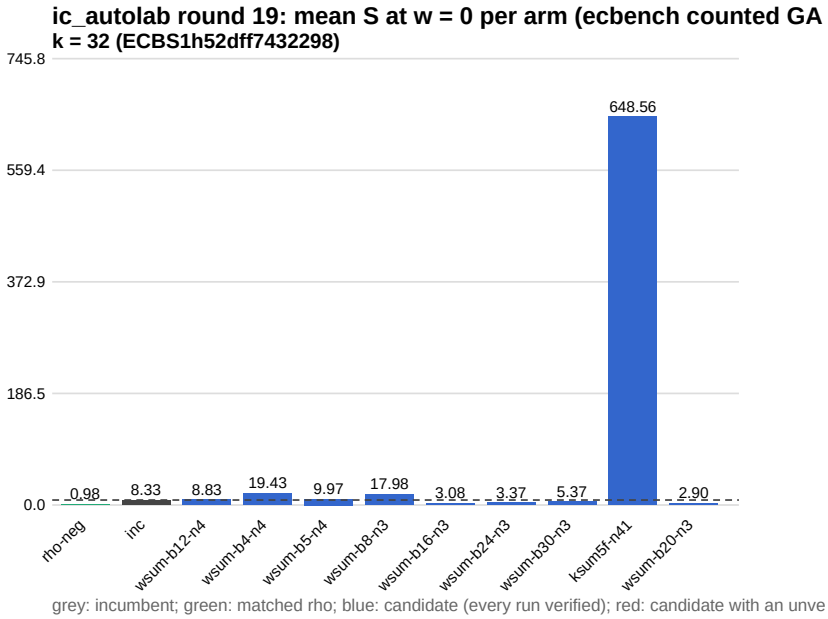
- k = 32: compare ops.status = ok, admissible = true; sel / inc = 0.3503 [0.329, 0.373] at w = 0; band: sel < inc; A/A inc-aa / inc interval [1.000, 1.000] contains 1

Train (development screen, in sample)

k = 32: session `ECBS1h52dff7432298` (run exit 0, verify exit 0, receipt sha256 `24d84aa535d58ba3d61c8f7043ee601ae1ccac0be4e951bbfd8b35628be09aac`)

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
`rho-neg`	rho	- -	`rho.negation`	32	32	0.982 / 1.079 / 1.948	matched reference
`inc`	ksum4	26 `prime-projected-abscissa:size=26,scan_cap=416`	`ksum:m=4`	32	32	8.328 / 8.849 / 13.539	incumbent
`wsum-b12-n4`	wsum-b12	4 `prime-projected-abscissa:size=4,scan_cap=64`	`wsum:coef=12,free=1`	32	32	8.826 / 8.892 / 9.485	cross-size transfer: wsum-b12-n3 is the k = 28 incumbent at N / N_a = 0.015
`wsum-b4-n4`	wsum-b4	4 `prime-projected-abscissa:size=4,scan_cap=64`	`wsum:coef=4,free=1`	32	32	19.427 / 23.259 / 57.745	cross-size transfer: wsum-b4-n4 is the k = 24 incumbent at N / N_a = 0.049
`wsum-b5-n4`	wsum-b5	4 `prime-projected-abscissa:size=4,scan_cap=64`	`wsum:coef=5,free=1`	32	32	9.972 / 11.879 / 29.047	cross-size transfer: wsum-b5-n3 is the k = 20 incumbent

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
							at N / N_a = 0.094
wsum-b8-n3	wsum-b8	3 prime-projected-abscissa:size=3,scan_cap=48	wsum:coef=8,free=1	32	32	17.984 / 21.534 / 53.487	exploration: family wsum-b8 untried at k = 32, N = 0.0001 x N_a
wsum-b16-n3	wsum-b16	3 prime-projected-abscissa:size=3,scan_cap=48	wsum:coef=16,free=1	32	32	3.081 / 3.527 / 7.542	exploration: family wsum-b16 untried at k = 32, N = 0.0001 x N_a
wsum-b24-n3	wsum-b24	3 prime-projected-abscissa:size=3,scan_cap=48	wsum:coef=24,free=1	32	32	3.369 / 3.521 / 4.895	exploration: family wsum-b24 untried at k = 32, N = 0.0001 x N_a
wsum-b30-n3	wsum-b30	3 prime-projected-abscissa:size=3,scan_cap=48	wsum:coef=30,free=1	32	32	5.367 / 5.447 / 6.172	exploration: family wsum-b30 untried at k = 32, N = 0.0001 x N_a
ksum5f-n41	ksum5f	41 prime-projected-abscissa:size=41,scan_cap=656	ksum:m=5,free=1	32	32	648.557 / 680.065 / 963.637	exploration: family ksum5f untried at k = 32, at its parent ksum5's best N = 41
wsum-b20-n3	wsum-b20	3 prime-projected-abscissa:size=3,scan_cap=48	wsum:coef=20,free=1	32	32	2.901 / 3.170 / 5.590	exploration: family wsum-b20 untried at k = 32, N = 0.0001 x N_a



train screen

Selection

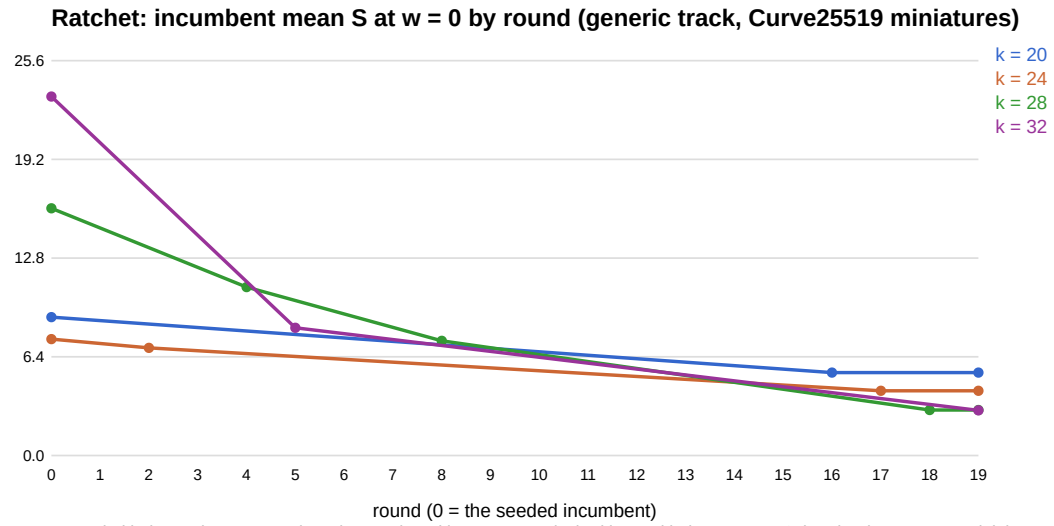
- k = 32: challenger `wsum-b20-n3` (Pareto alternative at w = 1: `wsum-b24-n3`)

Holdout sessions

k	session	run exit	verify exit	receipt sha256	levels
32	`ECBS1hf3547017358d`	0	0	`367ee066a71e7aade2557fe2633eb70b397be5ba91975b3e193e59951910a6e9`	{“verified”:480}

Ledger after this round

k	incumbent	since round	S (w = 0 / 0.1 / 1)	distance to rho
20	`wsum-b5-n3` (`prime-projected-abscissa:size=3,scan_cap=48` + `wsum:coef=5,free=1`)	16	5.384 / 5.609 / 7.638	3.6x / 3.5x / 3.0x
24	`wsum-b4-n4` (`prime-projected-abscissa:size=4,scan_cap=64` + `wsum:coef=4,free=1`)	17	4.201 / 4.433 / 6.528	3.4x / 3.3x / 2.8x
28	`wsum-b12-n3` (`prime-projected-abscissa:size=3,scan_cap=48` + `wsum:coef=12,free=1`)	18	2.949 / 3.222 / 5.676	3.0x / 3.0x / 3.0x
32	`wsum-b20-n3` (`prime-projected-abscissa:size=3,scan_cap=48` + `wsum:coef=20,free=1`)	19	2.937 / 3.213 / 5.697	3.1x / 3.1x / 3.1x



Recorded ledger values; a step down is a ratcheted improvement (paired interval below 1 at w = 0, band order at every weight).

ratchet

Evidence and deviations

- ecbench binary sha256: `e7242bc192bda8c8da9e64895afe943e8aa47cfa37bd490d656e2181758e9533`; `--cpus auto`, allow_busy = true; `verify --replay 8` per session; audit receipts in `analysis/`; sessions in `sessions/` (records in the artifact store when large).
- Route: the local route is an explicitly requested development screen, not the Iso Lab EKS default; counts are deterministic and replayed, wall time is not claimed. No row of this round enters the IC leaderboard or `benchmarks/README.md` (validation gate).
- Train selection is in sample (disclosed); the holdout is fresh (new `target_seed`, excluded exposures). Failed, timed-out and regressing arms are retained above.